

University of Sheffield

Currency and Control:

An Econometric Analysis of China's Dedollarisation efforts within the Petrodollar Context

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A report submitted in fulfilment of the requirements
for the degree of BSc in Politics, Philosophy and Economics

in the

School of Education

October 30, 2026

Abstract

China's role in the global economy has threatened the standing of U.S. Dollar hegemony. A central aspect to China's push for financial autonomy has been the dedollarisation agenda. One key case where this can be observed is in the energy markets where the petrodollar has sustained itself whilst aiding U.S. Dollar supremacy. Understanding shifts in the petrodollar system as a result of China's monetary policies or understanding how China reacts to the petrodollar is integral for evaluating possible future changes to the global political economy and its hierarchies. Much has been written in regard to China's internationalisation strategy, dedollarisation attempts and the intricacies of their energy sectors. However, little to no empirical analysis has considered the interplay of the three. This piece uses key macroeconomic monetary indicators to examine the structural relationships between China's FX reserves, U.S. exchange rate and Brent Crude oil prices. This is done with structural break detection, verified with ARIMA and VAR modelling and VAR-based Granger causality tests in an attempt to create a robust methodology to help understand exogenous elements and China's respective reactions. Results show that China's responses, in the oil and gas context, are largely reactive with more significant structural breaks taking places during period of geopolitical instability (2014 and 2022). Results also validate the significance of exogenous monetary shocks relating to the U.S. Dollar, confirming the academic consensus. Importantly, a key assumption is challenged by the results. Dedollarisation is arguably not a steady, linear and uniform movement as many academic may let on but, instead, often take the form of reactive, episodic 'recalibrations' according to energy dependencies and the impact of U.S. Dollar hegemony.

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Chapter 1

Introduction

Growing global rhetoric regarding self-sufficiency, resource nationalism and isolationist policy in the contemporary landscape are evident. Accordingly, the U.S. Dollar hegemony in international markets is an academic point of interest when considering the sensitivity of economies to exogenous aspects. The Dollar's role has been entrenched into the mechanisms of current global economies (Cohen, 2019; Gerding & Hartley, 2024). Specifically, pricing key commodities, providing a currency for global reserves systems and being the standard currency for international payments has cemented the U.S. Dollar's global has given the currency a resilience incomparable to that of any other major currency alternatives.

However, with the historically long trend of aggressive U.S. foreign monetary policy and aiding military occupation where it facilitates the interests of U.S. Dollar hegemony, it is expected that we see some countries shift towards other sources of monetary power (Zeidan, 2024; Eichengreen & Kawai, 2015). Primarily, the 'cold war' between the U.S. and China reflects a shift in global power dynamics. Whilst the U.S. Dollar remains indubitably supreme, the Chinese renminbi (RMB) has been internationalised in an attempt to hold its own against the dollar. Moreover, China's monetary policies reflect an overarching and somewhat regional attempt at dedollarisation. Dedollarisation, in this context, should be considered as a somewhat umbrella term, both refering to "efforts... to ascend the hierarchy by reducing structural dependencies on the dollar" (Katlenburner & Painceira, 2018) and, as Tooze (2021), notes, "Dedollarisation is as much a political act as it is an economic one"

1.1 The Petrodollar Background

This piece explores the Petrodollar in relation to growing sentiments of dedollarisation. As a key systemic form of U.S. Dollar dominance (Kilian, 2009; Farrell & Newman, 2019), the Petrodollar's impact should be somewhat empirically verifiable in China's given the Chinese government is, or at the least accused of being, at the forefront of dedollarisation efforts. Essentially, the argument is that China's dedollarisation strategy in the global oil markets is somewhat reactive and grounded in episodic, systemic reactions to exogenous shocks.

This argument will be put forward with a mixed-methods time series analysis. By looking

specifically at structural break points (Bai & Perron, 2003) we can diagnose the decisions made by the Chinese government in relation to monetary policy and dedollarisation. Analysis includes the USD/CNY exchange rate, foreign exchange reserves currencies and measures for internationalisation (RMB Tracker). Vector Autoregressions (VAR) and ARIMA models, used to verify the significance of the structural breaks and any possible interdependencies.

1.2 Paper Overview

The chapters in this piece are structured to reflect this layered inquiry. Chapter 2 reviews the theoretical and empirical literature on dollar hegemony, currency power, and dedollarisation. Chapter 3 discusses the paper's methodological approach and data sources. Chapter 4 presents the empirical results. Chapter 5 discusses the findings' implication and their relation to the existing literature, and Chapter 6 offers a culmination of this piece's results and discussion.

Chapter 2

Literature Review

This literature review builds on the foundational work of Cohen (2015) on 'currency power' and Norrlof's (2014) portrayal of a structuralist framework for US dollar hegemony. In his piece, Cohen outlines the currency hierarchy and how infrastructural advantages are used to maintain currency hegemony, specifically the US dollar. Drawing on Clark (2005) and Gelpern & Heller (2019), this piece will consider the 'soft law' agreements and how global financial power dynamics provide both the driving force and limiting factors for dedollarisation. As such, this piece aims to test the concept of dedollarisation against empirical evidence from the oil market. This is especially significant given the role of oil in global commodity markets weaponised to maintain US dollar hegemony. Dedollarisation policies on the matter effectively signify both symbolic defiance against a supremely dominant system and an attempt at restructuring currency power.

2.1 The Significance of Dedollarising the US Dollar/Oil Nexus

Since the 2007-2008 global financial crisis, the concept of dedollarization has become an increasingly attractive solution for dismantling US global financial dominance. The "exclusive position as the funding vehicle and pricing currency of the global oil trade" (Liu and Papa 2022), naturally, also belong to the US dollar. As such, the global oil and gas industries serve as a strong indication of the dollar's strength and signify their importance in the world economy. Namely, the 'West Texas Intermediate' (WTI) and 'Brent' are two of the world's leading crude oil pricing benchmarks, priced in dollar. Economists and academics' views on the matter generally share this sentiment, with Gal Luft (2018), the co-director of the Institute for the Analysis of Global Security, claiming that the "global commodity market, especially the USD1.7 trillion oil market" will be where the "future of the dollar will be decided". However, some scholars may hold that dedollarisation represents a symbolic movement away from US dollar Hegemony but caution overstating its significance given the importance of the US dollar in reserve and invoicing data (Eichengreen, Gerding, 2023)

Dating from 2007, scholars agree that policies involving the largest importers and exporters of energy globally (China and Russia, respectively), signify a real challenge to the US

petrodollar and infers an overarching 'currency war' (Liu & Papa, 2022; Tshikovhi et al., 2025; Song, 2007). The policies range from direct challenges to Dollar hegemony, to indirect focuses on domestic currency usage, discussed in detail later in this piece. Other than the clear objective of strengthening the prospects of a renminbi dominated oil trade and therefore the strength of their currency overall, China's endeavour also attempts to dismantle the power of US sanctions on the global markets. Incidentally, it is these same sanctions placed on major oil exporting countries, like Iran and Russia, that fuels the sentiments towards a 'nondollar alternative payment system' in the global political landscape. However, empirical studies point towards the unchanging nature of US dollar hegemony up until late 2023 (Gerding Hartley 2024), highlighting the robustness of the petrodollar system.

2.2 The growing Allure of the Chinese Yuan

Nonetheless, policies from BRICS (Brazil, Russia, India, China and South America) and China's economic infrastructure as well as other dedollarisation-related foreign policy initiatives all contribute to a vigorously systemic execution of dedollarisation. Relative to previous challenges, especially in the middle east, BRICS' attempts may pose a more realistic and difficultly thwarted threat to the petrodollar and US dollar hegemony. Programs such as the Belt and Road Initiative, centred around yuan trading in over 150 countries and international organizations has seen success and significant backing. Indeed, it is "widely viewed as one of the largest infrastructure investment projects ever launched" (Mathews and Selden 2018). On the same note, the Asian Infrastructure Investment Bank (AIIB) and the BRICS New Development Bank (NDB) also provides global incentives to trading in the Yuan.

Critical to this conversation is the fact that, historically, the US has been maintaining control over the petrodollar through military occupation and economic sanctions (Shipley, 2007). The US dollar has essentially evolved into an instrument of unilateralism in order to maintain the US' place in the world economy (Abbas, Kibria, Arif, and Ali 2025). Tyler Shipley shares this view, essentially claiming "the primary motivating factor" of involvements in oil-rich countries, such as Iraq and Libya were maintaining the status quo for the "manner in which oil is sold... and, most importantly, how they pay for it" (Shipley, 2007). In reality, the petrodollar system itself serves as a reminder of US power in foreign affairs and, therefore, it is unsurprising why countries are increasingly opting in favour of the Yuan in this context, essentially standing against US dollar hegemony through the global oil markets. The petrodollar represents more than the functioning of the global oil trade but, rather, a reflection of deep-rooted structural power and the perpetuation of global power dynamics through "monetary capabilities, capital markets, and politico-military force" (Norrlof, 2014)

2.3 The Renminbi as a growing and adapting response to the Petrodollar

As Shipley (2007) explains, "controlling the flow of oil can be compared to a modern version of controlling strategic shipping lanes". This is the academic consensus with another political scientist, Carla Norrlof (2014), claiming that it is the control of the oil-currency nexus that paves the way for "prestige, seignorage, balance of payments flexibility, and policy autonomy" for the US economy. Notably, however, in 2017, China surpassed the US as the largest importer of crude oil (EIA 2018), This, coupled with the BRICS-backed Chinese and Russian 'yuan-oil futures' contract settled at the Shanghai International Energy Exchange in 2018, signifies a serious and growing threat to the petrodollar system. The 'yuan-oil futures' was quick to garner contracts, accounting for 14% of global contracts by mid-2018, simultaneously eroding the WTI and Brent (Mathews and Selden 2018). Iran, Iraq, Venezuela, Indonesia and Angola followed suit, opting to use Yuan as a payment option, to various degrees. Crucially, in March 2024, The Wall Street Journal reported that Saudi Arabia may be considering to trade futures with the Chinese Yuan. If this were to occur, it would signify a momentous shift especially given almost all oil trade has been conducted with the US dollar since 1974 with the help of Saudi Arabia. Adding to the attractiveness of the 'yuan-oil futures', is its gold convertibility which provides security for benefactors and appeal for the Asian and African markets where gold reserves are more prominent (Mathews and Selden, 2018). As Mathews and Selden (2018) clarify, this marks China stepping away from their focus on financial liberalisation and Special Drawing Rights (SDRs) as a medium for dedollarisation and a step towards bilateral trade agreements that solidifies their position in the volatile global economic climate.

SDR is an 'international reserve asset' that's value is determined by five currencies (US dollar, Euro, Yuan, Yen and the UK pound), with China's currency being added in 2016. This reflected China's aggressive internationalisation attempts for their currency at the time, with the aggressive, reformist strategy for the renminbi being initiated in the early 2010s. Eichengreen and Xia (2018) essentially break down the strategy as an attempt from the Chinese government to "force the pace of domestic financial liberalization". However, the Yuan's inclusion in the SDR proved problematic. One condition for the inclusion was that the renminbi be deemed 'freely usable', though some academics and political commentators (Cohen 2017) question if this condition was ever met and if China should have been ejected from the SDR long ago. Nevertheless, China's attempts for a 'freely usable' Yuan backfired, being described by Eichengreen and Xia (2018) as "an impatient reformers' Trojan horse". A 'trojan horse' in the sense that the Chinese government wanted rapid international acceptance of the renminbi, but their financial systems (FX, regulations and banking) could not support vulnerabilities (primarily the relatively massive capital outflows).

With loosened capital controls, and the domestic economic and political context contradicting the push for financial liberalisation, the 'speculative attack' of 2015-16 proved extremely problematic for the Chinese economy. The Mundell-Fleming trilemma (the impossible trinity

theory) states that is impossible for a country's economy to have the trifecta of a fixed exchange rate, free capital mobility and independent monetary policy (Flemming 1962). China violated all three, striving for control over domestic interest rates and monetary policy, a relatively stable Yuan and, most critical to the matter, capital influx to internationalise the renminbi. As Zeiden (2024) outlines, it was the close to a trillion-dollar capital flight fuelled by households and domestic firms alike that shocked China's expectations and overwhelmed authorities. In reaction, 'the people's bank of China' limited mergers and acquisitions, devalued the Yuan in a managed float and receded to a to stricter capital control. Since 2016, China has pivoted from seeking a more liberalised and flexible capital market to "developing cross-border yuan-denominated trade settlement systems" (Zeiden, 2024) with politically aligned countries. Though now aiming to maintain capital control for financial stability, China's overarching de-dollarisation objective remains a central aspect. Rather than systematically displacing the US dollar hegemony, China's strategy, as of 2016, can be better understood as reinforcing the barriers against US sanctions. Zeiden (2024) agrees, describing de-dollarization as, "in essence, a risk management measure for China" and an attempt to support a financial infrastructure with less vulnerability to US leverage amid a time of intensifying global economic conflict.

This piece will examine two key areas: dedollarisation effects post-2016 and an overarching analysis of China's economic sensitivity to the oil market dating from 2006. By looking at timed policy implementations by the leading challenger of US dollar hegemony (China), we can better understand the development of dedollarisation attempts in the oil context and its effectiveness, specifically after the arguable failure of China's related policy attempts pre-2016.

2.4 Empirical Studies

Emprical studies linking the concept of dedollarisation, specifically, to the oil markets are minimal, and even more so when applied to BRICS and China centred dedollarisation policies. Nonetheless, quantitative analysis, though fragmented, does provide interrogation of the interplay between monetary policies, oil and gas commodity pricing and macroeconomic indicators' volatility despite no discussion of dedollarisation as a named concept. On this, this piece intends to fill the gap in the literature, testing the concept of dedollarisation against empirical findings. In particular, structural vector autoregression (SVAR), Bayesian VAR (BVAR), and mixed-level modelling are all used in the surrounding literature and influenced the makeup of the methodology used in this piece. However, as SVAR requires restricting variables by asserting the assumption of immediate causal relationships, and this piece intends to prove the existence or non-existence of these relationships, it is not applicable. As for Bayesian VAR, the time series range in this piece is large enough not to warrant BVAR use. As such, standard VAR and MVAR is used (one bivariate and one multivariate (MVAR)).

The following empirical findings build on a developed understanding of oil shocks, their cause and their general effect. Kilian (2008) and Peersman & Van Robays (2012) emphasize

the need to disaggregate oil shocks by the source (supply, induced by exogenous shocks, or speculative demand) to interpret outcomes with more understanding of the mechanisms at play. When applied to dedollarization contexts, this differentiation becomes essential. Dedollarisation policies such as those aimed at preventing economies from facing dollar-driven oil price cycles may backfire if the transmission mechanisms of oil market shocks are negligently overlooked.

Cross and Nyugen (2016) examine the effects of global oil price shocks to the growth of China's economy. Using BVAR, their findings suggest that generic demand shocks (induced by, say, strong global economic activity requiring oil such as a booming global market) produced a positive effect on China's economy (measured through real GDP). On the other hand, when an oil shock is not supplemented with an increase in global economic activity, such as a speculative demand shock induced by geopolitical risk, the result is a depression in China's real GDP. On the same note, supply shocks to the oil market also yielded the same results, namely a decline in real GDP. This reaffirms the aforementioned discussion on China's vulnerability to US dollar hegemony. Despite initiatives intended to avert exposure to the petrodollar system, China's sensitivity to dollar-denominated price volatility of oil remains a problematic reality.

The claim that China is a dominant player in determining oil price volatility is somewhat reflected in a VAR analysis by Zhang and Choi (2013). The model found secondary statistically significant results, showing China's contribution in global oil consumption had swayed world oil prices. Unsurprisingly, OPEC supply shocks and the previously mentioned speculative demand shocks were shown to be the primary drivers of price movement. Unconventionally, using a Directed Acyclic Graph gave the authors the ability to assess how countries' domestic oil prices shifted others given 'contemporaneous innovation transmissions' (market shocks). The results confirm that OPEC and US' policies in regard to oil heavily fuelled global market volatility, with China's market activity following. Essentially, though being the largest importer of oil, China is not a price setter.

Gao (2023) shows that the USD/CNY exchange rates would fluctuate in the short-term following "divergence by the two central banks", though this had "no long-term fundamental changes" on the exchange rate. Similarly, Gerding and Hartley (2024) report the resilience of US petrodollar dominance especially in oil invoicing, despite the attempts of systemic change for the past two decades and aggressive US foreign policy utilising the petrodollar. In their piece, they build on and update previous metrics initially introduced by Maggiori (2019). By looking at currency invoicing, denomination of global debt, central bank currency reserves and foreign exchange (FX) transactions, the authors reaffirmed the inertia of US dollar hegemony. The piece also found that the "yuan-based futures" was failing to displace WTI and Brent benchmarks significantly. This boiled down to US dollar hegemony over oil trades and debt and FX reserves being generally denominated in US dollar. On the other hand, Metin-Özcan & Us (2007), utilising an impulse response function (IRF) analysis, claims that "the variance decomposition suggest that dollarization is still persistent, lending support to the notion of hysteresis." What he means here is that the context and past conditions of

global markets, through heavy reliance on the dollar, effect present systems even though US dollar hegemony maybe weakened. Essentially, path dependence on a global financial system, dollarisation, is trapped into equilibrium, uneasily revertible and influenced.

As a culmination of the available literature, it is clear that the empirical evidence clashes. Whilst the measurable impacts on US dollar hegemony and the petrodollar specifically remain ambiguous, dedollarisation attempts especially in the global oil market remain ambitious and spearheaded by China. Naturally, the inertia of the established, robust and resilient US petrodollar framework puts forward methodological challenges and complexities to the matter.

Where this piece will differentiate itself from previous literature is in its handling of oil revenue stability as a dependent variable following dedollarisation policies. Instead of looking to global pricing or other general macroeconomic indicators, this piece looks to examine the global oil trade as medium for dedollarisation. Moreover, structural break analysis at points of momentous dedollarising policy shifts, such as the introduction of the yuan-oil futures contract and the Yuan's inclusion in the SDR, links the concept of dedollarisation to specifically timed empirically observable shocks rather than treating dedollarisation as a 'latent-variable', as many analyses of the subject-matter do (e.g., Gao 2023, Gerding & Hartley 2024). The SWIFT RMB tracker essentially enables this, by providing data to indirectly measure the effectiveness of dedollarsation through promotion of domestic currency.

Chapter 3

Methodology

3.1 Data Sources and Treatments

The research presented in this study was based on multiple data sources, listed as follows. World Bank for the USD/CNY exchange rate. Specifically, the Global Economic Monitor (GEM) database was used. SWIFT RMB Tracker Reports was used to get information regarding renminbi (RMB) transactions. Federal Bank of St Louis for China's foreign exchange (FX) reserves, excluding gold. US Energy Information Administration (EIA) for brent crude oil prices.

While USD/CNY exchange rate data and Brent Oil prices were reported daily, monthly means were taken and used for this piece to cohere with the literature. Missing figures were scarce, but any gaps found were filled with linear interpolation.

As for the exclusion of gold from the foreign exchange variable, this was done deliberately given gold's role in international markets. Firstly, as a "safe haven" currency (Baur & Lucey, 2010), gold reserves are unlike currency reserves oftentimes stored by governments as a longer-term store of value. As a non-fiat currency, surrounding literature tends to exclude gold in empirical studies regarding dollarisation, FX reserve volatility and currency substitution (Eichengreen, 2011; IMF, 2021). Secondly, gold fluctuates somewhat independently of any other reserves (Baur & Lucey, 2010; Dempster & Artigas, 2010), being more dependent on interest rates than other assets and currency exchange figures. The contrast in the determinants of gold prices compared to other assets is made especially clear in times of geopolitical crisis, where gold is significantly less volatile (Baur & Lucey, 2010; Dempster & Artigas, 2010). When real interest rates are low or negative, gold is demanded in greater quantities, given its minimal yield loss (Baur & McDermott, 2010; Erb & Harvey, 2013). Essentially, the weak links to the Petro-dollar, the energy market, generally, or to the US dollar's strength in international markets justify the exclusion.

3.2 Statistical Tools: VAR Models

Two vector autoregressive (VAR) models were used in this study. VAR (Vector Autoregressive) models are systems of equations where each variable is a linear function of past logs of itself, along with the other variables in the system (Enders, 2015). These models are especially effective when attempting to capture the dynamic interrelationships of multiple variables.

The two VAR models used in this dissertation have a distinct time range. The first VAR model used in this research spans from June 2009, when the intergovernmental organisation BRICS was officially formed, until 2024, therefore containing the period after the global financial crisis and the COVID pandemic. The second model has a more limited time period due to the SWIFT RMB tracker only operating from the end of 2014 (November) onwards.

VAR lag selection tests (LL, LR, FPE, AIC and HQIC tests) were used on both VAR models used to determine the lag selection of each model.

In order to examine the significance of lagged variables in predicting other variables in the VAR models, Granger-causality tests were conducted utilizing both OLS regression and VAR. Granger-causality tests essentially compare regression models of a variable (Y) with and without the addition of another variable (X). If the addition of another variable (X) provides statistically significant improvements to the model, then we can conclude that X can be used to 'granger-cause' Y. OLS regression Granger tests are limited to two variables. Moreover, enders (2015) and Lütkepohl (2005) note, granger causality tests using VAR models are more effecive given the models are shown to mitigate omitted bias and the inclusion of lags betters the model stability. Given this piece's VAR models satisfy these conditions Granger-Causality tests using VAR models were presented.

VAR models require the assumption that the data involved be stationary. According to Gujarati and Porter (2009) a time series is said to be stationary "if its statistical properties such as mean, variance, and autocovariance are all constant over time". Nonstationary data can lead to false regressions reflecting invalid inferences. When VAR analysis is used with non-stationary time series data it can therefore lead to pseudo-vector autoregressions based on invalid inference. To avoid the circumstance of pseudo-vector autoregression, the Augmented Dickey-Fuller (ADF) test was used.

ADF tests examine the presence of unit roots in the data, that is when the autoregressive parameter of a time series is equal to one. This would mean that shocks to the variables would have a permanent effect (Gujarati & Porter, 2009) and that subsequent analysis would provide misleading results. The null hypothesis of ADF tests is that the series has a unit root and is therefore non-stationary. Therefore, statistical significance of the ADF test (which amounts to rejecting the null hypothesis) would support that the time series data used in the model is stationary, as required by VAR. ADF-testing of all variables proved that the null hypothesis could not be rejected. This implied that the series had an unit root (the variables where initially non-stationary). To fix this issue logs and first differencing of each variable had to be computed. After the latter, the produced p-values indicated stationarity was achieved. Therefore, the log and first differences of each variable were the specific form of the data used throughout the analyses.

3.3 VAR models

3.3.1 VAR model 1

The first VAR model looks at the relationship between brent crude oil prices (POil), the CNY/USD exchange rate (Xcny usd),and FX reserves (excluding gold) (RFX) as endogenous variables. Exogenous variables (X) include selected break points, discussed in the next section, and energy outputs. This model intends to examine the interplay between these factors to determine China's economy's sensitivity to exogenous dollar-related shocks. As noted above, the first VAR model in this study spans from June 2009, when the BRICS was first founded, until December 2024 and provides an overarching analysis of China's sensitivity to oil prices over the last decade.

As Zhao (2023) highlights, currency exchange markets respond quickly to changes in crude oil prices and, therefore, instead of using standard VAR lag selection tests listed in the subsection above, Eigenvalue stability conditions of different VAR models were compared to assist in determining lag selection. These stability conditions are based on whether eigenvalues fall inside a unit circle (their absolute value being less than one). If they do not fall inside a unit circle, then the model should be considered 'unstable', to a degree, or possessing unit roots.

$$P_t^{\text{oil}} = a_1 + \sum_{p=1}^{3} b_{11}^{(p)} P_{t-p}^{\text{oil}} + \sum_{p=1}^{3} b_{12}^{(p)} X_{t-p}^{\text{cny usd}} + \sum_{p=1}^{3} b_{13}^{(p)} R_{t-p}^{\text{FX}} + C_1 X_t + \varepsilon_t^{\text{oil}},$$

$$X_t^{\text{cny usd}} = a_2 + \sum_{p=1}^{3} b_{21}^{(p)} P_{t-p}^{\text{oil}} + \sum_{p=1}^{3} b_{22}^{(p)} X_{t-p}^{\text{cny usd}} + \sum_{p=1}^{3} b_{23}^{(p)} R_{t-p}^{\text{FX}} + C_2 X_t + \varepsilon_t^{\text{cny usd}},$$

$$R_t^{\text{FX}} = a_3 + \sum_{p=1}^{3} b_{31}^{(p)} P_{t-p}^{\text{oil}} + \sum_{p=1}^{3} b_{32}^{(p)} X_{t-p}^{\text{cny usd}} + \sum_{p=1}^{3} b_{33}^{(p)} R_{t-p}^{\text{FX}} + C_3 X_t + \varepsilon_t^{\text{FX}}.$$

$$\begin{bmatrix} P_t^{\text{oil}} \\ X_t^{\text{cny_usd}} \\ R_t^{\text{FX}} \end{bmatrix} = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} + \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \\ b_{31} & b_{32} \end{bmatrix} \begin{bmatrix} P_{t-1}^{\text{oil}} \\ X_{t-1}^{\text{cny_usd}} \\ R_{t-1}^{\text{FX}} \end{bmatrix} + \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \\ b_{31} & b_{32} \end{bmatrix} \begin{bmatrix} P_{t-2}^{\text{oil}} \\ X_{t-2}^{\text{cny_usd}} \\ R_{t-2}^{\text{FX}} \end{bmatrix} + \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \\ b_{31} & b_{32} \end{bmatrix} \begin{bmatrix} P_{t-3}^{\text{oil}} \\ X_{t-3}^{\text{cny_usd}} \\ R_{t-3}^{\text{FX}} \end{bmatrix} + \begin{bmatrix} C_1 X_t \\ C_2 X_t \\ C_3 X_t \end{bmatrix} + \begin{bmatrix} \varepsilon_t^{\text{oil}} \\ \varepsilon_t^{\text{cny_usd}} \\ \varepsilon_t^{\text{FX}} \end{bmatrix}$$

3.3.2 VAR model 2

The second VAR model specifically looks at the RMB tracker (RMBTTracker) and China's reliance on crude oil (through oil refining data: POil). The RMB tracker is not exactly a measure of global trades in Yuan. Rather, it measures the volumes of transactions flowing

through interbank and cross-border payment systems tracking customers' international wire credit transfers and cross-border bank-to-bank transfers, such as a credit transfer transmitted from the sender's (debitor) bank to the recipient's (creditor) bank abroad. The VAR 2 model used is therefore an attempt to relate dedollarisation attempts (reflected by the amount of global trade conducted in the Chinese Yuan instead of the US dollar) with China's push against the petrodollar. This model also includes numerous exogenous structural break variables (X) to examine China's Renminbi's reaction to geo-political and world economic events not too dissimilar to work done by Kilian (2009) and (Stock & Watson 2001).

$$\mathrm{RMBT}_t^{\mathrm{Tracker}} \;=\; a_1 \;+\; \sum_{p=1}^{2} b_{11}\,\mathrm{RMBT}_{t-1}^{\mathrm{Tracker}} \;+\; \sum_{p=1}^{2} b_{12}\,P_{t-1}^{\mathrm{Oil}} \;+\; y_1^{\mathrm{BreakDummies}}\,X_t \;+\; \varepsilon_t^{\mathrm{Tracker}},$$

$$P_t^{\mathrm{Oil}} \;=\; a_2 \;+\; \sum_{p=1}^{2} b_{21}\,P_{t-1}^{\mathrm{Oil}} \;+\; \sum_{p=1}^{2} b_{22}\,\mathrm{RMBT}_{t-1}^{\mathrm{Tracker}} \;+\; y_2^{\mathrm{BreakDummies}}\,X_t \;+\; \varepsilon_t^{\mathrm{Oil}}.$$

$$\begin{bmatrix} RMBT_t^{\mathrm{Tracker}} \\ P_t^{\mathrm{Oil}} \end{bmatrix} = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} + \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} \begin{bmatrix} RMBT_{t-1}^{\mathrm{Tracker}} \\ P_{t-1}^{\mathrm{Oil}} \end{bmatrix} + \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} \begin{bmatrix} RMBT_{t-2}^{\mathrm{Tracker}} \\ P_{t-2}^{\mathrm{Oil}} \end{bmatrix}$$
$$+ \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} \begin{bmatrix} RMBT_{t-3}^{\mathrm{Tracker}} \\ P_{t-3}^{\mathrm{Oil}} \end{bmatrix} + \begin{bmatrix} y_1^{\mathrm{BreakDummies}} X_t \\ y_2^{\mathrm{BreakDummies}} X_t \end{bmatrix} + \begin{bmatrix} \varepsilon_t^{\mathrm{oil}} \\ \varepsilon_t^{\mathrm{FTSE}} \end{bmatrix}.$$

3.4 Test for Unknown Breaks and ARIMA models

To determine the break points of interest and see where the Renminbi may be directly affected by fluctuations in global energy prices, following Bai and perron (2003), a sequential test for unknown break points (estat sbcusum) was constructed for the USD/CNY exchange rate, the FX reserves (excluding gold) and Brent crude oil prices, as presented in tables 1-3. These results were nonetheless not strictly relied upon. Where unknown breaks correlated with recent timings of relevant geopolitical events, surrounding literature and contextual factors inferred the chosen structural breaks. Given their ability to specifically examine structural changes in time series forecasting, and to further explore VAR models' finding, ARIMA models were also employed. ARIMA models consist of autoregressive (AR), integrated (I), and moving average (MA) components (Gujarati & Porter, 2009). The AR component examines the relationship between current values of the variables in a series against previous values whilst the MA component focuses on the relationship between current values and previous error terms. The integrated (I) element differences the variables to achieve stationarity (Box & Jenkins, 1976). The use of ARIMA models is commonplace in surrounding literature, specifically when dealing with econometric time series data, and they were included in

the analysis to help validate the results, provide a generally more robust analysis, and highlight any discrepancies when compared against the VAR models. By including the chosen structural break point dummy variables discussed in the next section, the ARIMA models are expectantly more robust and better able to capture the nuance and interplay between the variables, given structural shocks and change (Enders, 2015; Gujarati & Porter, 2009). In order to provide a metric for model performance and confirm the models' parsimony (Gujarati & Porter, 2009), the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) were utilized in selecting the ideal ARIMA specification.

Chapter 4

Results And Analysis

4.1 Break Points

4.1.1 Analysis of Unknown Break Tests

In the following section, the results of the break tests mentioned in the methodology will be analysed and discussed. The 'likely determinants of the breaks' based on the literature surrounding the topic are discussed and detailed below. However, consider that the economic variables involved in this analysis are affected by countless exogenous factors that are impossible to account for in entirety. This casts doubt over establishing definitive causal relationships in the subject matter. (Chen, Rogoff & Rossi, 2010), Therefore, the 'likely determinants of the breaks' should only be considered as partial, possible causes.

4.1.2 CNY/USD Exchange Rate

4.1.2.1 CNY/USD Exchange Rate with Breaks

Table 4.1: (CNY/USD)

Test	**Test statistic**	**1% Critical Value**	**5% Critical Value**	**10% Critical Value**
$F(1 \mid 0)$	2.92	12.29	8.58	7.04
$F(2 \mid 1)$	3.58	13.89	10.13	8.51
$F(3 \mid 2)$	7.36	14.80	11.14	9.41
$F(4 \mid 3)$	16.08	15.28	11.83	11.04
$F(5 \mid 4)$	13.18	15.76	12.25	10.58

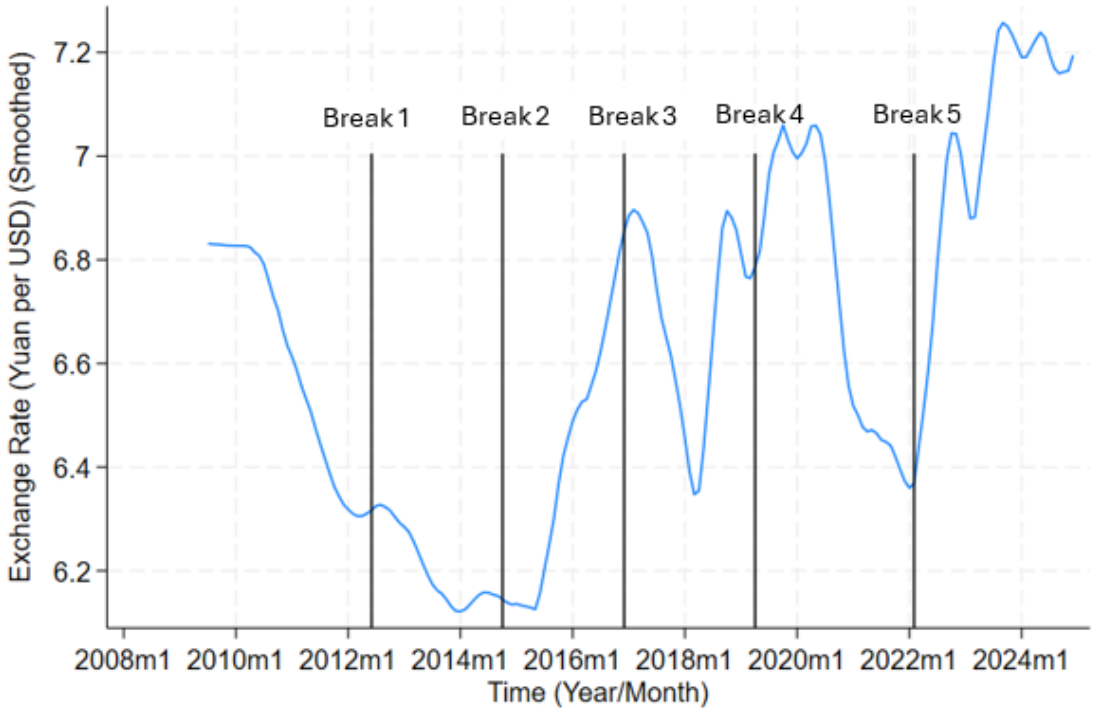


Figure 4.1: CNY/USD exchange rate from 2008 through 2024

4.1.2.2 Break 1: June 2012

This is likely a market reaction to an attempt to pursue exchange rate flexibility from the People's Bank of China (PBOC) through the expansion of the daily trading band of the Yuan from ±0.5% to ±1% in April 2012. This marked a step towards market-based price setting of exchange rates, internationalisation and a relaxation of the Chinese currency's dependence on the US dollar (Subramanian & Kessler, 2012).

Observed through initial market volatility, market players accordingly positioned themselves in anticipation of a more globally rooted Yuan. The policy shift signalled a dedollarisation attempt to promote confidence in the renminbi as a more globally prominent and possible reserve currency. Today, significantly more cross-border settlements are conducted in Yuan and scholars such as Subramanian & Kessler (2012) attribute this change, which is interpreted in our analysis as a break in the time-series, to the relatively and increasingly stronger position of the Renminbi on the world stage.

One of the numerous pieces of empirical evidence for this would be the aforementioned RMB Tracker's values. The RMB accounted for 0.34% of total global payments in January 2012, before the break. In the following year and after the break (January 2013), this figure rose to 0.63% (Swift, 2013). Given the relatively short timeframe, this economic achievement provides significant insight into China's dedollarisation capabilities, namely its effectiveness.

4.1.2.3 Break 2: September 2014

Exchange rate pressure from a sharply appreciated U.S. dollar and the slowing economic growth of China set the stage for the 2015 devaluation of the Yuan. Expected interest rate hikes by the US Federal Reserve created downward pressure on prominent alternative currencies (Eichengreen & Kawai, 2015) due to a shift towards assets traded in the dollar.

The shifting dynamics of exogenous factors caused market predictions and appropriate pricing shifts (IMF, 2015). Market predictions indicated that investors, guided by analysts' expectations, anticipated the PBoC to respond to the appreciating U.S. Dollar with cautious views about the Chinese economy in the short term. Specifically, further exchange mechanism loosening from the PBoC or relevant yuan stimulating policy were expected (Zhang & He, 2017). This expectation came to fruition.

According to the IMF, (2015) "The PBoC reduced benchmark interest rates in November 2014... The reserve requirement ratio was also lowered multiple times." These were the first cuts to the benchmark interest rate since 2012. Importantly, China's economic outlook during the same period of this break was weak, reporting 7.4% growth, their lowest in 24 years according to China's National Bureau of Statistics (Reuters, 2015). Amid growing concerns, the influx of liquidity into the economy due to an increased amount of loans (from the benchmark interest rate reduction), resulted in a clear need for an exchange rate reform. This was predicted by academics such as Subacchi & Huang (2012). According to the latter, this marked a need to strengthen "the convertibility of the RMB and the liberalization of the capital account, which, in their turn, depend on the reform of the financial and banking sector."

4.1.2.4 Break 3: December 2016

This break represents the strict capital controls that followed the capital outflow crisis and subsequent Yuan devaluation in 2016. Moreover, it happened amid President Donald Trump's first presidential term in the U.S. which also led to downward pressures on the Yuan via the strong U.S. dollar (Zhang & He, 2017). China's FX reserves also fell below 3 trillion U.S. dollars, prompting the shift to stricter FX transactions and capital outflows (IMF, 2017; BIS, 2017) . Essentially, this period depicted the transition of China's exchange rate mechanism to a managed, floating system (Eichengreen & Kawai, 2015).

4.1.2.5 Break 4: September 2019

In August of 2019 the PBoC reactively allowed the USD/CNY rate to exceed a milestone of 7.0, a level not seen in a decade, following a U.S. announcement of 300-billion-dollar tariffs on Chinese goods (IMF, 2020). China was named a "currency manipulator" drawing attention to China's monetary policy practices and strategy (U.S. Treasury, 2019). This marked a time of heightening tensions and increased international attention to China, as an economic spearhead of dedollarisation efforts with Yu (2020) describing China as "at the forefront of efforts to challenge the dominance of the US dollar".

4.1.2.6 Break 5: February 2022

Amid weakening economic growth due to pandemic-related capital inflow issues and the Russia-Ukraine war, the Chinese managed float currency's heightened sensitivity to external shocks was exposed (World Bank, 2022; IMF, 2022). The aggressive appreciation of the U.S. dollar due to raised interest rates by the U.S. Federal Reserve added to downward pressure on the Yuan. The CNY/USD, once again, rose above the 7.0 level, inferring the volatility of the Yuan, despite a cautious monetary policy strategy for the managed float exchange rate system (IMF, 2022; BIS, 2022).

4.1.3 FX Reserves excluding gold holdings.

Table 4.2: FX Reserves (excluding gold)

Test	**Test statistic**	**1% Critical Value**	**5% Critical Value**	**10% Critical Value**
F(1|0)	19.23	12.29	8.58	7.04
F(2|1)	9.57	14.80	11.14	9.41
F(3|2)	9.57	14.80	11.14	9.14
F(4|3)	11.54	15.28	11.83	10.04
F(5|4)	12.23	15.76	12.25	10.58

Note: Rounded to 2 decimal places
Note: Suggested months: 2017m2, 2022m9 (only two breaks were detected)

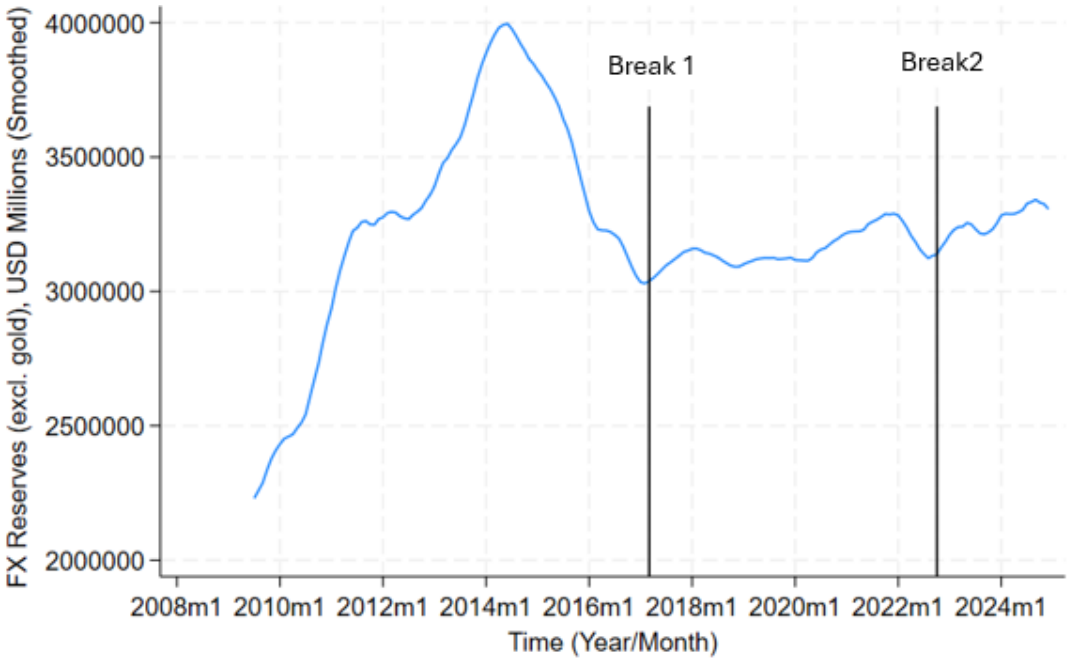


Figure 4.2: FX Reserves (exc gold) with Structural Breaks

4.1.3.1 Break 1: February 2017

For added context regarding this break, look to 4.1.2.3 and 4.1.2.4 which detail China's exchange rate reform. As SAFE (2016) and BIS (2017) report, the capital flight crisis resulted in a symbolic fall for foreign exchange reserves below three trillion U.S. dollars. The exchange rate liberalisation provided security and less volatility for China's foreign exchange reserves. Mainly, the central daily reference for the USD/CNY exchange rate was no longer set by the PBoC but rather based on the previous day's closing rate (Zhang & He 2017). This lifted significant pressure off China's FX reserves, as inferred by the graph above.

China's FX reserves had already been on a negative trend given heavy intervention during the 2015 capital flight crisis. By January 2016 FX reserves had fallen below a symbolic 3 trillion U.S. dollars (IMF, 2017; BIS, 2017). As aforementioned, this led to tightened capital and monetary controls. Notably, this marked the end of the generally downward trend and a step towards a more stable and robust renminbi (Ghosh et al., 2016).

4.1.3.2 Break 2: September 2022

Whilst the Yuan depreciated past 7.0 in the CNY/USD exchange rate, FX reserves remained somewhat nominally stable. Nevertheless, this marked a change in strategy for the FX reserves system for China. Importantly, and on a related note, it marked increased investments by China in gold (World Gold Council, 2022; BIS, 2022). This is the most likely reason for the break placement.

4.1.4 Brent Oil

Table 4.3: Brent Prices

Test	Test statistic	1% Critical Value	5% Critical Value	10% Critical Value
F(1|0)	8.71	12.29	8.58	‘7.04
F(2|1)	3.07	13.89	10.13	8.51
F(3|2)	2.23	14.80	11.14	9.41
F(4|3)	8.09	15.28	11.83	10.04
F(5|4)	6.99	15.76	12.25	10.58

Note: Rounded to 2 decimal places
Note: Suggested months: 2016m1 (Only one break detected)

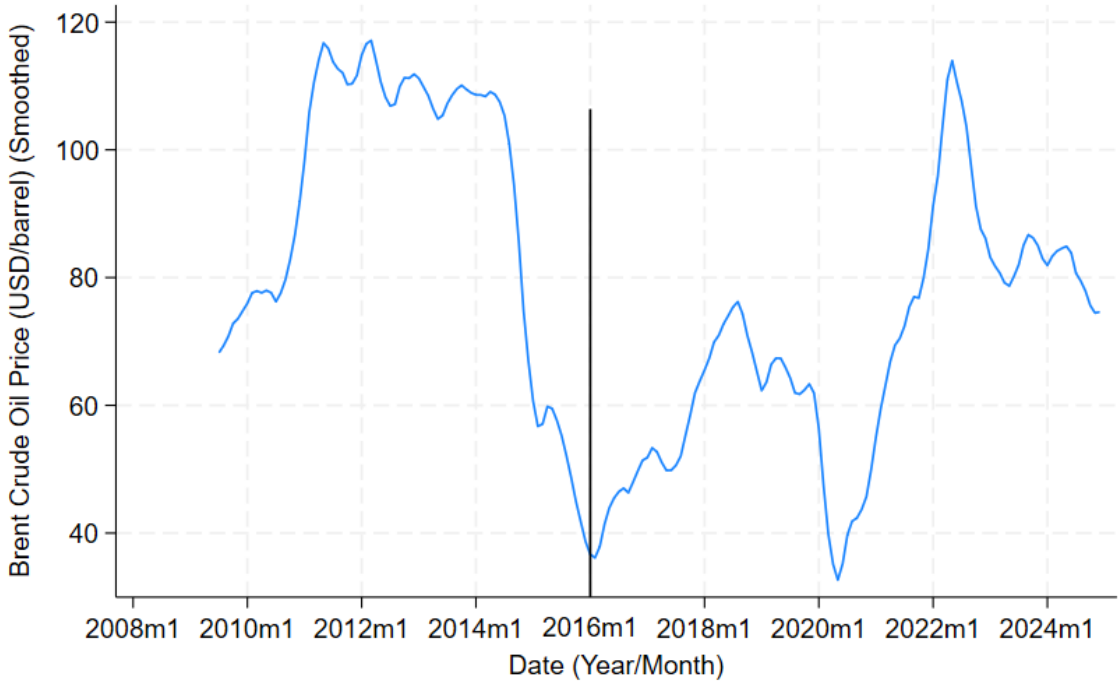


Figure 4.3: Brent Oil Prices with Structural Breaks

4.1.4.1 Break 1 January 2016:

This break represents increased downward pressure on crude oil prices and an eventual global oil market collapse. More specifically, this represented the peek of a collapse originating in mid 2014, as shown in Figure 4.3. The clear consideration to make is that this coincided

with China's aforementioned economic reform. The low prices for oil, scholars claim (Arezki & Blanchard, 2014), reduced pressure on China's current account and the Yuan given that China is a net oil importer. Brent crude oil prices fell to $30.70 per barrel from $38.01 in December 2015. Whilst the low oil prices are usually associated with less pressure on the current account and a stronger purchasing power for oil importing countries, China's economic and exchange rate reform somewhat mitigated any significant alleviations.

The oil market collapse originally (2014) resulted in deflationary pressures onto oil importing countries, like China. The subsequent a decrease in associated costs of production led to stagnating rates of global inflation (Arezki & Blanchard, 2014). At this time, heightened capital outflows in China and slowing 'momentum' in commodity exporting countries, globally ((BIS, 2016; IMF, 2015b), reflected a volatile state for China's economy. This break represents a culmination of exogenous factors that have clashing yet unique impacts on China. Both domestic economic reform and exogenous dollar-hegemony related variables provide polarizing pressure on China's outlook.

4.2 Tests for known breaks on break Locations:

Given that we have somewhat determined the possible 'breaks' of interest and described some of the significant events that could be associated to them, multiple statistics and stability condition checks were conducted on several regressions. We begin presenting results from regressing the variable FX Reserves or Crude Oil prices (see Table 4.4 below). While the regression involved Chow-tests, as per the academic norm when testing for the validity of structural breaks, the former were purposefully omitted. As Bai & Perron (2003) suggest, the Chow Test is "unsuitable when multiple structural changes are present". Moreover, the more complex the model, the higher the likelihood in discrepancies from Chow Test results.

Table 4.4: Tests for structural breaks

Break	F-Statistic for Model	p-value for F-Test	Wald Test Chi-Square	p-value for Wald Test
USD/CNY Exchange Rate				
Break 1 – June 2012	3.00	0.085*	2.999	0.083*
Break 2 – September 2014	3.780	0.053*	3.777	0.052*
Break 3 – December 2016	3.793	0.043**	0.191	0.062*
Break 4 – September 2019	0.190	0.662	0.016	0.898
Break 5 – February 2022	5.440	0.021**	5.445	0.020**
FX Reserves (excluding gold)				
Break 1 – February 2017	3.120	0.079*	3.152	0.076*
Break 2 – September 2022	0.190	0.662	0.176	0.6746
Brent Prices				
Break 1 – January 2016	0.760	0.3839	0.761	0.382

Note: Results rounded to 3 Decimal Places

4.3 ADF tests

This section details the augmented Dickey-Fuller tests which indicate stationarity (see methodology section). Following this point, logged and first differenced variables were used (relevant for ARIMA modelling where integrated values remain at 0 given stationarity is achieved). Results for the Brent crude oil monthly variable vary amongst the two VAR models, due to the differing scope of the time series.

Table 4.5: ADF Tests on VAR 1 Variables

Variables:	**t-statistic:**	**p-values:**
CNY/USD	-0.753	0.833
FX reserves	-3.817	0.000***
Brent crude oil	-1.784	0.390
Energy output	-1.930	0.318
Logged and first differenced		
CNY/USD	-8.265	0.000***
Brent crude oil	-10.527	0.000***
Energy output	-1.930	0.000***

*Note: * Indicates statistical significance at the 10% level. ** Indicates significance at the 5% level and *** at the 1% level*
Note: Results are rounded to 3 Decimal Places

Table 4.6: ADF Tests on VAR 2 Variables

Variables:	**t-statistic:**	**p-values:**
RMBT Tracker	-1.698	0.432
Brent crude oil	-1.885	0.339
Logged and first differenced		
RMBT Tracker	-8.265	0.000***
Brent crude oil	-7.932	0.000***

*Note: * Indicates statistical significance at the 10% level. ** Indicates significance at the 5% level and *** at the 1% level*
Note: Results are rounded to 3 Decimal Places

4.4 VAR lag selection

Higher values of Log Likelihood (LL) indicate a better fit to the model (Lütkepohl, 2005). Likelihood Ratio (LR) results report whether additional lags improve the model's fit (Greene, 2018). DF refers to the degrees of freedom included in estimations (Greene, 2018) and LL results are provided with a p-value. FPE (Akaike's Final Prediction Error) is a forecast related measure of model fit, lower values reflect a better fit (Lütkepohl, 2005). The Akaike Information Criterion (AIC) (Akaike, 1974; Enders 2015), HQIC Hannan–Quinn Information Criterion (HQIC)(Hannan & Quinn ,1979) and SBIC Schwarz Bayesian Information Criterion (SBIC) (Schwarz ,1978; Lütkepohl, 2005) all infer model 'complexity', with lower values reflecting 'more complex' and robust models.

Table 4.7: VAR Model 1 identification

Lag	LL	LR	df	p-value (LR)	FPE	AIC	HQIC	SBIC
0	-1624.83				697.624	17.899	17.928	17.970
1	-1210.21	829.23	16	**0.000**	8.735	- 13.519	13.662	13.871*
2	-1175.57	69.289	16	**0.000**	7.118	-13.314	13.571*	13.948
3	-1153.34	44.449*	16	**0.000**	6.652*	-13.246*	- 13.617	14.161
4	-1142.4	21.877	16	0.147	7.041	13.786	-9.3841	14.498

*Note: * Indicates the recommended lag length for each test. Figures were rounded to 3 Decimal Places*

Table 4.8: VAR Model 2 identification

Lag	LL	LR	df	p-value (LR)	FPE	AIC	HQIC	SBIC
0	368.839				3.4e-06	-6.922	17.928	-6.871
1	388.893	40.108	4	**0.000**	2.5e-06	-7.224	-7.1633	-7.074
2	401.419	25.052*	4	**0.000**	2.1e-06*	-7.385*	7.283*	-7.134*
3	404.071	5.303	4	0.258	2.2e-06	-7.360	-7.217	-7.008
4	407.650	7.158	4	0.128	2.2e-06	-7.352	-7.169	-6.900

*Note: * Indicates the recommended lag length for each test. Figures were rounded to 3 Decimal Places*

Markedly, the p-values for lags 1-3 in VAR 1 and 1-2 in VAR 2 report 0.000 indicating that the first three lags strongly reject the null. Insofar that inclusion of lags 1-3 and 1-2 in the two models, respectively, significantly improve the VAR model.

4.5 VAR Granger-Causality Tests

Table 4.9: Granger-causality with VAR 1

Direction of Causality	Chi-Squared	Lags	p-value
Brent Oil Prices → CNY/USD	10.06	3	0.018**
Brent Oil Prices → FX Reserves	2.03	3	0.57
CNY/USD → Brent Oil Prices	2.74	3	0.43
CNY/USD → FX Reserves	1.86	3	0.05**
FX Reserves → CNY/USD	1.79	3	0.43
FX Reserves → Brent Oil Prices	2.79	3	0.62

*Note: * indicates a significance level below 10% and ** below 5%. Results rounded to 2 decimal places.*

Table 4.10: Granger-causality with VAR 2

Direction of Causality	Chi-Squared	Lags	p-value
RMB Tracker → Brent Oil Prices	10.06	2	0.364
Brent Oil Prices → RMB Tracker	2.03	2	0.996

*Note: * indicates a significance level below 10% and ** below 5%. Results rounded to 2 decimal places.*

4.5.1 ARIMA modelling

ARIMA modelling was conducted on variables included in the first VAR model to examine the validity of the proposed breaks. Before incorporating breaks into the possible ARIMA models, model specifications were determined, to begin with, using partial-autocorrelation and autocorrelation plots for AR (Auto-Regressive) and MA (Moving Average) inputs, respectively. As shown below, only the first lags of the USD/CNY exchange rate seem especially significant. Following the plots, ARIMA (1,0,1), was taken into primary consideration.

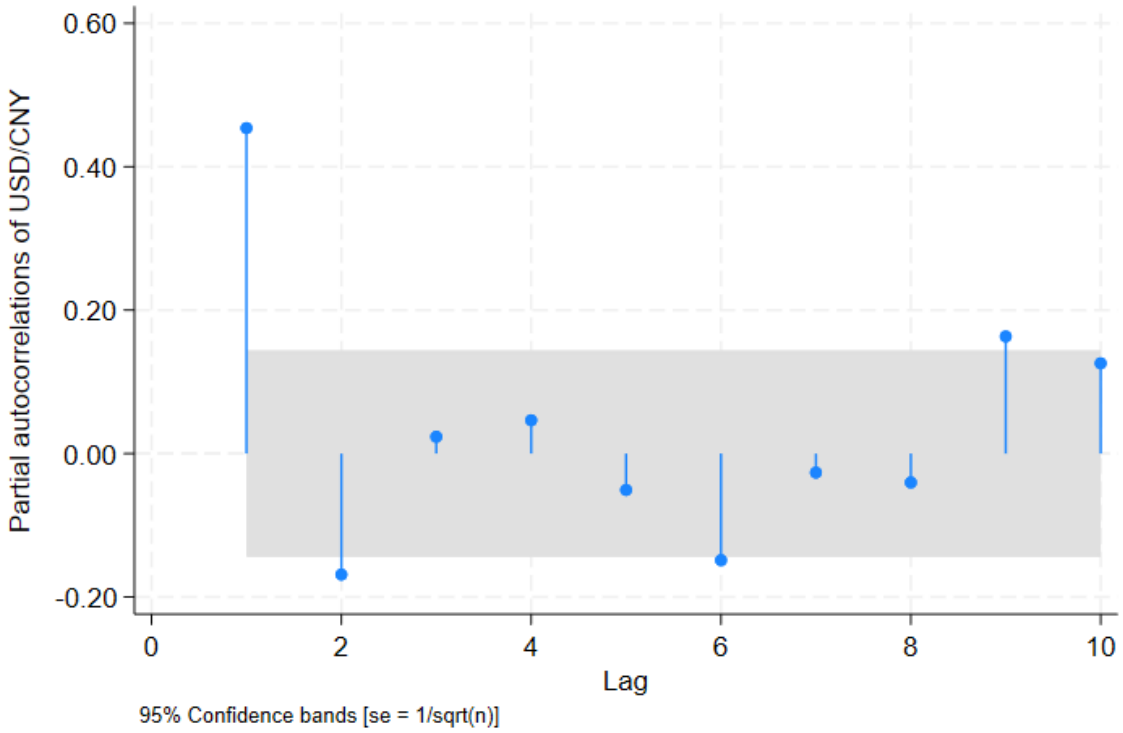


Figure 4.4: Autocorrelation Plot

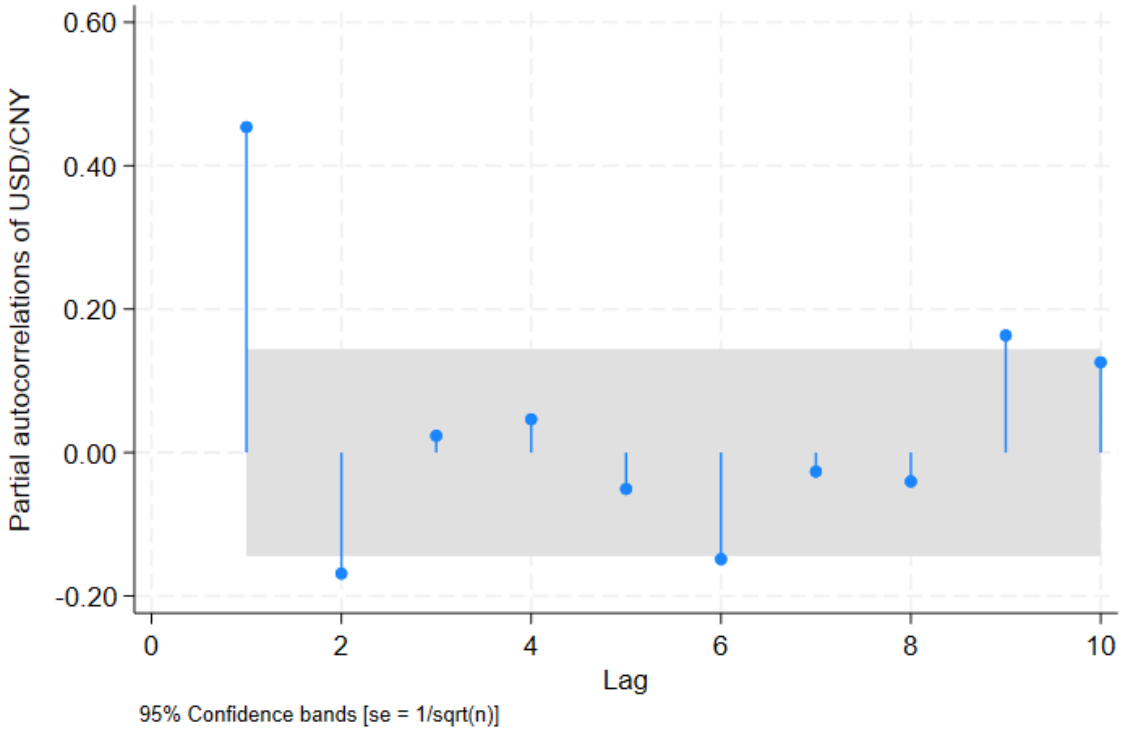


Figure 4.5: Partial - Autocorrelation Plot

4.5.1.1 ARIMA Model comparison

Despite autocorrelation and partial autocorrelation plots providing sufficient evidence to adopt an ARIMA (1,0,1) model, several other models were tested, and their BIC and AIC values are reported below. Akaike Information Criteria (AIC) and Bayesian Information Criteria are used to test a models fit and 'parsimony'. By testing for these criterions, an opportunity to adopt a 'better', more parsimonious model may present itself. Following testing, Model 4 (0,1,1) was adopted (with the lowest AIC and BIC scores being attributed to the 'most parsimonious' of the options).

Table 4.11: ARIMA model selection for USD/CNY

Model Selection Criteria	**Model 1**	**Model 2**	**Model 3**	**Model 4**	**Best Model**
ARMA Specification	(1,1,1)	(2,1,1)	(1,1,0)	(0,1,1)	-
Log likelihood	615.95	616.19	613.19	614.84	Model 4
AIC	-1223.91	-1222.38	-1221.06	-1223.69	Model 4
BIC	-1211.00	-1206.25	-1211.38	-1214.01	Model 4
Best Model	-	-	-	-	**Model 4**

Note: Results rounded to 2 Decimal Places

4.5.1.2 ARIMA Modelling with Breaks

The significance of AIC and BIC are discussed in 4.3

Table 4.12: AIC and BIC tests of ARIMA 4 Models

Model Description	**AIC**	**BIC**
ARIMA model 4 (0,1,1)	-1223,69	-1211.01
Model 4 with USD/CNY Break 1 -	-1223.40	-1210.97
Model 4 with USD/CNY Break 2 -	-1223.87*	-1208.94
Model 4 with USD/CNY Break 3 -	-1223.05	-1210.15
Model 4 with USD/CNY Break 5 -	-1223.687*	-1211.01
Model 4 with FX Reserve Break 1 -	-1221.758	-1208.855

*Note: Results rounded to 2 Decimal Places. *Symbolises an improvement to the model.*

4.6 VAR 2 RMBT Break Analysis

Initial break testing for VAR 2, using estat sbscum on a simple regression proved not to reject the null hypothesis (that there are breaks) (Bai, Perron, 2003; Gujrati, Porter, 2009). Hence, the original assumption was that there were no breaks to be observed. Using xtbreak, the test for unknown breaks, revealed another break in February 2020 with a 95% confidence interval (and was therefore included in the following individual break testing. Further break analysis included using break results from VAR 1 that tested with significance alongside the RMB tacker variable. Individually testing each break provided the following results.

Table 4.13: Break Analysis of RMBT

Break Timing	**Coefficient**	**p-value**	**F-statistic**	**p-value (F-test)**	**Conclusion**
January 2016	-0.00755	0.850	0.040	0.846	Non-signfiicant
February 2017	0.0265	0.350	0.880	0.350	Non-signfiicant
September 2022	0.0650	0.049*	3.910	0.049*	Significant
December 2016	-0.0170	0.559	0.340	0.559	Non-signfiicant
February 2020	0.0430	0.163	1.980	0.163	Non-Significant

Chapter 5

Discussion

This section will interpret and discuss the findings of Chapter 4 withing the overarching context of China's dedollarisation efforts. One key portion of the previous chapter interrogates the significance of the proposed break point locations of the two VAR time series. The other aspect highlights the interrelationships of the discussed variables (see methodology section).

By looking at the structural breaks of the USD/CNY exchange rate, FX reserves (excluding gold) and Brent crude oil prices, individually, this paper isolates structural economic changes in China within the context of dedollarisation efforts. The paper used both sequential (Bai, Perron, 2003) break and known break tests and then incorporated significant breaks into VAR and ARIMA models to determine significance of variables involved. The results generally reaffirm the academic consensus, that dollar hegemony remains constant. Surrounding literature's discussion on the institutional advantages of the Dollar system (Gerding & Hartley, 2024; Eichengreen & Gerding, 2023) and the lack of comparable structural scale or associated trust in currency alternatives (Cohen, 2019; Abbas et al., 2025; Acharya, 2014) points to a U.S. Dollar which is heavily resistant to dedollarisation efforts, at least in the context of China. Moreover, the analysis reveals that dollar supremacy is challenged by policy changes and strategic shifts from China, particularly in managing exchange rate reform and FX exchange rate policy in relation to the global oil market. This reinforces the discussion by Subramanian and Kessler (2012) who point to an emerging but gradual rise of the renminbi associated with a sustained monetary rebalancing. As Kesler notes, "In East Asia, there is already a RMB bloc, because the RMB has become the dominant reference currency, eclipsing the US dollar, which is a historic development. (2012) "Nevertheless, this is met with notable and volatile relationship to US monetary policy changes (Cohen, 2019; Zhang & He, 2017).

The significance of global oil prices to the Chinese economy is predictable given China's status as the largest importer of oil. Gao (2023) has also previously observed China's trade balance, inflation and exchange rate in the context of oil price volatility. This piece builds on the sentiments of other quantitative evidence on the subject. By observing policy changes around moments of global oil market disruption, we can understand the interconnectedness of China's dedollarisation efforts and reactionary policy shifts, reflecting Kilian's (2009) argument that global oil price shocks can generate significant, asymmetric and permanent

macroeconomic effects.

China's growing concern towards their domestic economy in the last decades is essential to this paper's analysis. The RMB internationalisation strategy and FX reserve changes aligns with Gerding and Hartley's (2024) position that oil-dollar dominance remains central to U.S. dollar hegemony. More concerningly, Hartley (2024) also claims that challenges to this system face structural inertia. This paper's empirical results somewhat reaffirm this claim. As such, China can be said to have been actively attempting to reshape the global monetary system, evidenced by the RMB Tracker but this study's empirical findings contextualises this, finding that the global oil market continues to restrict the pace and volatility of dedollarisation efforts (Zeidan, 2024; Yu, 2020).

Granger causality tests revealed statistically significant causality running from Brent oil prices to the USD/CNY exchange rate, reaffirming the findings of Cross and Nguyen (2016), who used BVAR models to show that oil price shocks substantially influence China's GDP and exchange rate movements.

5.1 Significance of Structural Breaks

In 4.1 this paper discussed the suggested break points and their supposed relevance. Table 4.4, which presents the test for known breaks, interrogates whether these break points hold significance within their models, according to methodology from Bai & Perron (2003). The estat sbcusum test for VAR 1 and VAR 2 variables generally failed to reject the null of break stability, while the sequential xtbreak test detected five breaks), highlighting the superior sensitivity of the Bai & Perron method for time-series with multiple unknown shifts. This result aligns with Gujarati and Porter (2009) caution against relying solely on recursive stability tests when regime shifts are expected.

The implications of these tests are as such:

5.1.1 USD/CNY exchange rate and RMB Tracker

Breaks 1 (June 2012) and 2 (September 2014) proved significant at 10% and 5% levels, respectively. These breaks are associated with the PBoC's widening the yuan trading band (break 1) and the effects of a strongly appreciating U.S. Dollar (break 2).

In June 2012, the yuan's daily trading band increased from $\pm 0.5\%$ to $\pm 1\%$, representing a commitment to a more market orientated system with currency internationalisation at its heart. This incentivised foreign trade partners and commercial investors to consider the Renminbi more seriously. The success of this change is made evident by this analysis.

The significance of break 2 illustrates the sensitivity of China's monetary system to external shocks. Essentially, US Federal Reserve policy shows how the persistently asymmetric power of the U.S. Dollar remains dominant despite the rise of China in the world economy, as a global superpower with the second largest economy globally, set on dedollarisation. This means China's policy makers must consider global dollar cycles despite its constant strategical shifts away from the dollar or towards strengthening the Yuan.

Break 3 (December 2016) reporting high significance was expected given the change to a manged float system. Break 4 (September 2019) was nonetheless insignificant. This can be tied to rising Chinese-U.S. relations. As predicted, this break was likely insignificant due to anticipatory pricing by market forces.

Break 5 (February 2022) confirms the Renminbi's sensitivity to exogenous, geopolitical shocks like the Russia-Ukraine war and aggressive interest rate hikes in the U.S. Interestingly, break analysis of the RMB Tracker (Table 4.13) showed statistical significance during September 2022 too. Though expected, China's financial payment flows, and relevance of the Yuan, thrived. This is most likely attributed to the decline and restrictions placed on the Russian economy, at least in the short term, that then had longer-term, positive effects for China's economy, trade balance and dedollarisation efforts. Primarily, Russian imports from China rose in an attempt to circumvent G7 sanctions. A significant portion of these imports were conducted in Yuan. These trade agreements solidify themselves over time due to the continued sanctions on Russia. Overall, the later contextualises the significance of this break 5.

5.1.2 FX Reserves (Excluding Gold)

Only the first break appeared to be significant (February 2017). This illustrates how capital control and closing rates post 2016 marked a significant shift in China's monetary strategy.

5.1.3 Brent Crude Oil Prices

The only break point proved to be insignificant. This represented a historical collapse of the oil market and the plummet of oil prices. The insignificance of this break is surprising yet it relates to the consistency and elasticity of the global oil market.

Essentially these results paint the picture of a reactionary Chinese economy that implements dedollarisation efforts episodically. Meaning, anti U.S. Dollar domination sentiments are not linear and gradual, but sudden and responsive. Specifically, China appears to gear its monetary policy towards dedollarisation in times of financial market instability. (Subramanian & Kessler, 2012; Eichengreen & Kawai, 2015; Zhang & He, 2017).

5.2 ARIMA Model Comparison and Validation

Despite autocorrelation plot indications of an AR(1) structure, the ARIMA model (0,1,1), was selected for its superior model fit as indicated by the lowest AIC (-1223.69) and BIC (-1214.01) values (Table 4.11). This choice aligns with Box & Jenkins (1976) model parsimony criteria. Importantly, the inclusion of break dummies in ARIMA models generally improved fit, with the September 2014 break yielding the lowest AIC (-1223.87), These results validate the ARIMA model's utility in confirming structural changes inferred from VAR-based diagnostics. The contrast between break-inclusive and baseline ARIMA models illustrates how accounting for structural shocks provides a more accurate forecast of RMB exchange behavior. As such,

the ARIMA analysis substantiates the VAR break findings in dedollarisation-linked model interpretation and adds a robustness check to a break analysis in the time-series.

5.3 VAR Lag Structure and Model Fit

VAR Model 1 and VAR Model 2 lag selection processes yielded optimal lags of 1-3 and 1-2, respectively, based on the lowest AIC, HQIC, and SBIC values (Table 4.7 and Table 4.8). These results indicate temporal inertia in the transmission of oil and currency shocks across China's macrofinancial variables. In both models, early lags show the strongest influence, confirming short-term dependencies.

5.4 Granger Causality Interpretation (Tables 4.9 and 4.10)

As a reminder, a variable is said to 'granger-cause' another variable "if information about the past of the first variable helps predict the future of the second variable better than using only its own past information" (Clarke, Granato, 2005)

The Granger causality results for VAR Model 1 confirms that Brent oil prices significantly Granger-cause the CNY/USD exchange rate ($p = 0.018$). This finding aligns with literature linking oil price shocks and the USD to emerging market exchange volatility (Kilian, 2009). The directionality from oil to CNY suggests that energy import costs, largely denominated in USD, continue to drive RMB fluctuations. This highlights and explains why the U.S. Dollar-Oil nexus is so problematic, and would be taken into heavy consideration, for China's monetary policy.

Additionally, CNY/USD was found to Granger-cause FX reserves ($p = 0.05$), revealing that exchange rate behaviour is an active determinant of FX reserve accumulation (excluding gold), possibly due to the PBOC's intervention strategies.

However, all causality relationships in VAR 2 (RMBT and Brent) were statistically insignificant. While this weakens the empirical case for energy-dedollarisation linkages through RMB trade settlement volumes, it may also reflect measurement limitations of the RMB Tracker or the lag structure used. A more probable alternative which is inferred by the analysis of this piece is that dedollarisation may be unfolding more through institutional reforms rather than immediate changes to transactional flows, in China's case specifically.

5.5 Contextualizing Findings

The petrodollar system remains a key foundation of dollar hegemony. China, accordingly, appears aware of its limitations and sensitivities in regard to dedollairising the global energy markets. As shown by causality results of VAR 2, China's ambitions must lie elsewhere. Specifically, China's resources have pivoted to AI-critical minerals, technology and minerals (IEA, 2021; Kiggins, 2022) where China prioritises the renminbi through yuan-denominated underlying financial systems. China's economy and foreign policy surrounding these sectors

signifies a new 'Cold War' where China is seen to be laying the structural foundations for dedollarisation through infrastructure development and then leveraging this against the Dollar's influence (Greene, 2024; Zeidan, 2024). As such, influence over commodity pricing and relevant global trade ramifications, such as the petrodollar system, may not represent monetary power. Instead, somewhat intangible control over resources key for future economic growth reflects China's clash with the U.S. Dollar.

The sentiments of this paper's finding correlate heavily with aforementioned literature surrounding the subject. Cohen (2015) and Norlof's (2014) theoretical foundations are affirmed, U.S. Dollar dominance is apparent. Gerding and Hartley's (2024) work provides a key motive for this piece. In their analysis, Renminbi-denominated oil futures, BRICS trade agreements, similar symbolic shifts, were found not to significantly impact the robustness of U.S. Dollar hegemony. Vulnerabilities exposed by sensitivities to exogenous geo-political variables and the capital flight crisis (Liu & Papa, 2022; Abbas et al., 2025), caused a clear shift away from the previous SDR liberalization goals (Eichengreen & Xia, 2018). This also corresponds with Zeiden's (2024) recent analysis of Chinese dedollarisation attempts as a risk management response.

5.6 Validating Breaks through ARIMA Modeling

Table 4.12 demonstrates that the inclusion of some of the determined structural breaks improved the ARIMA model fit according to the AIC/BIC criterion. Specifically, Break 2 in the USD/CNY time series (September 2014) yielded the best model fit (AIC=-1223,87)

Notably, other break dummies yielded significantly lesser or obscure improvements to the model fit. The reasons for the lack of improvement to the model are somewhat obscure. This may point to the limits of isolated policy change in the face of a structurally entrenched dollar hegemony (Gerding & Hartley, 2024). Another possible explanation is reinforced by the academic narrative from writers like Gujarati & Porter's (2009) that would assert that only certain break timing can materially shift the structure's dynamics, and the relevant variables.

As per Table 4.11, ARIMA (0,1,1) as the most parsimonious mode, should be treated as the benchmark for further pieces that scrutinise the relationship between the USD/CNY exchange rate and the petrodollar system. Moreover, this points to a gap in the surrounding literature. Specifically, many quantitative pieces may undervalue the significance of accounting for China's episodic policy shifts through structural breaks in the time series which, otherwise, may not be captured.

5.7 Methodological Constraints

This piece offers empirical evidence for China's dedollarisation strategies within the petrodollar background. However, the focus on institutional reform and informal trends in China's monetary practices may overshadow nuanced shifts beyond the time series. These would

not necessarily be covered by VAR or ARIMA modelling with structural break analysis. Additionally, break analysis depends on the validity of chosen breaks. The determinants for these breaks in this paper, as discussed, are largely based on inferences from the surrounding literature. Conflating points of structural breaks (such as that commonly seen with the Russia-Ukraine war and Corona-Virus affect overlaps) is a considerable issue. Notably, as has been shown by Gujarati and Porter (2009), economic shifts may not translate into statistical significance. Measurement limitations are evident in this analysis. Other than the limited time frame of RMB Tracker, the data lacks nuance on trade dynamics, contract types etc. Similarly with Brent oil prices, other alternative pricing benchmarks are not considered and may provide an additional layer of investigation. Finally, these results may not clearly reflect China's spearheading role in global dedollarisation efforts. China's diversification into growing industries may paint a clearer picture (Zeidan, 2024; Greene, 2024).

5.8 Summary

The discussion has interpreted the statistical findings in the broader context of dedollarisation. Significant breaks in 2014 and 2022 corroborate shifts in China's currency regime. The Granger-causal role of Brent oil prices on RMB volatility challenges the autonomy claim of the Yuan. Still, improvements in model fit upon the inclusion of break dummies suggest dedollarisation efforts remain substantive and statistically significant.

Ultimately, the quantitative evidence points to moderate short-term volatility mitigation primarily through reactionary monetary policy and sustained increasing relevance of the RMB. Nevertheless, the structural weight of the petrodollar and the institutional nature of dedollarisation require ongoing empirical reassessment.

Chapter 6

Final Remarks

This piece has explored China's dedollarisation attempts within the lens of the oil and gas markets, considering the structural implications of the petrodollar. As a culmination of sequential and known break tests, ARIMA model analysis, and VAR-based Granger causality tests, this piece confirms the previous academic sentiment regarding the prevalence of dollar hegemony. However, a key insight this analysis unveils is a trend among the structural breaks and their determinants. China's efforts in the oil and gas markets to undermine the effects of the U.S. dollar have been mostly reactive. Indeed, this is reinforced by rising and ongoing anti U.S. sentiment as a reaction to technological restrictions (Minghao, 2023), resource and technological trade wars and geopolitical militaristic tensions (Zhang, 2020; Acharya, 2014). Moreover, given that empirical evidence points to generally unsuccessful attempts to trump the petrodollar, it is unsurprising that China adopts an aggressive and proactive stance on the matter. Specifically, the Breaks thoroughly discussed at in 2014 and 2022 reflects China's economy's sensitivity to the U.S. dollar whilst affirming the narrative that dedollarisation does not take the form of linear progression, but, instead, somewhat reactive monetary and political adjustments.

Whilst this piece has concluded a reactive stance from China regarding the petrodollar, this does not necessarily reflect a lack of proactive Chinese dedollarisation policies. A RMB bloc in East Asia, for example, represents one of the most substantive examples of dedollarisation with significant regional monetary and political implications. The 'shift to the east' from regions like the Middle East, Africa and East Asia is not coincidental. It is the result of over a trillion U.S. Dollar investment that promotes cross border financial agreements centred around China and the Renminbi (Hurley, Morris & Portelance, 2018; Yu, 2020). Understanding these shifting relationships requires acknowledgment that China is offering state-driven, secure investment with less political and economic ramifications than that of the IMF and the U.S. (Sun, 2022).

Another especially relevant tool to diagnosing China's relationship to the petrodollar is ,what some academics may label (Allison, 2017; Brands, 2020), the 'cold war' over global mineral markets and technology, such as AI development. This supports the argument that China is simply funnelling more resources into the sectors of their economy that can uphold

their self-reliant stance on Dollar hegemony. Monetary sovereignty, as we see it changing today, is stepping towards 'computational resources', not just oil and gas flows (Farrell & Newman, 2019; Lee, 2021). If the petrodollar appears to be untouchable, then perhaps China's domination over Rare Earth Elements (REEs), which are essential components of growing sectors in global technology markets (IEA, 2021; Kiggins, 2022), will provide China with 'strategic resilience' (Minghao, 2023) and a medium for dedollarisation.

The finding of this study regarding the episodic and reactive monetary polices of China is a small insight into China's broader dedollarisation strategy. By constructing alternative financial infrastructure to combat Dollar hegemony whilst simultaneously reacting accordingly to sustained U.S. monetary pressure from mediums like the petrodollar, China is striving to secure its place atop the world economy (Allison, 2017; Jacques 2012). Pioneering and controlling the chains of supply for desired future technologies only aids this narrative. As such, the empirical analysis provided does not point to China's resignation against the petrodollar but an attempt to maintain monetary sovereignty through currency internationalisation.

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Appendices

Stata Commands and Scripts

Example of filling in missing data:

```
ipolate CNYUSD numdate, gen (CNYUSD)
epolate
```

Converting to weekly data:

```
collapse (mean) (variables) by
(monthly)
```

Setting monthly data for time series analysis:

```
tsset numdate monthly
```

Generating log of variables:

```
gen l_CNYUSD = log(CNYUSD)
```

Generating first differences of variable:

```
gen d_l_CNYUSD = D1.l_CNYUSD
```

Testing for unknown break points:

```
xtbreak d_l_CNYUSD inter trend,
vce(hac)
```

Augmented dickey fuller test:

```
dfuller (variable)
```

Example for testing for lag selection for VAR:

```
varsoc var1 var2...
```

Example of VAR regression:

```
var d_l_CNYUSD d_l_Brent,
exog(break1) lag(1/3)
```

Testing for VAR stability conditions:

```
varstable
```

VAR granger-causality tests:

```
vargranger
```

Generating dummy break variable:

```
gen break = 1 if trend == 169 |
trend > 169
```

Chow test:

```
reg d_l_CNYUSD d_l_dcoilbrenteu break
1 break1dummy
test break1 break1dummy
```

Test for known structural break point:

```
estat sbscum
```

Test for known structural break at Oct 2022:

```
estat sbknown, break(tm(2022m10))
```

Partial autocorrelation/autocorrelation tests:

```
pac d_l_CNYUSD
ac d_l_CNYUSD
corrgram d_l_CNYUSD
```

Constructing ARIMA model:

```
arima d_l_CNYUSD, arima(1,0,1)
```

Granger-causality tests with OLS:

```
regress d_l_CNYUSD L(1/4).d_l_CNYUSD
L(1/4).d_l_dcoilbrenteu
Test L1.d_l_dcoilbrenteu
L2.d_l_dcoilbrenteu
L3.d_l_dcoilbrenteu
L4.d_l_dcoilbrenteu
```